Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

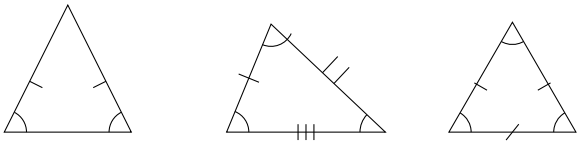
Section 1: Types of Plane Shapes and Their Properties

Plane shapes, also called two-dimensional (2-D) shapes, are flat shapes that lie completely in one plane. They include the triangle, the rectangle, the square, the parallelogram, the trapezium, the circle and all the other polygons.

Let's begin with the triangle, a closed shape with three straight sides. Triangles are named by how many of their sides are equal, as fig-01 shows. Look at the little ticks on the sides: sides with the same number of ticks are equal in length.

• An isosceles triangle has two equal sides (one tick on each of the two slanted sides). Its base angles are equal, and it has one line of symmetry.
• A scalene triangle has no equal sides (one, two and three ticks on its three sides). No angle is equal to another, and it has no line of symmetry.
• An equilateral triangle has three equal sides (one tick on each of the three sides). All its angles are equal, each measuring 60°, and it has three lines of symmetry.

Visual



Isosceles Scalene Equilateral

Isosceles: a single tick on each of the two slanted sides, an arc at each base angle. Scalene: one, two and three ticks on the three sides, an arc at each vertex. Equilateral: a single tick on each of the three sides, an arc at each vertex.

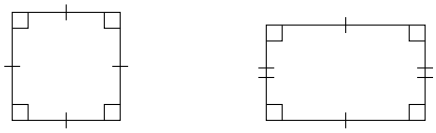
Next come the square and the rectangle, shown in fig-02. The right-angle marks at each corner tell you that every corner is a 90° angle.

The square:
• all four sides are equal,
• it has two pairs of parallel sides,
• all its angles are equal (90° each),
• it has four lines of symmetry,
• its diagonals bisect it at a right angle.

The rectangle:
• only opposite sides are equal (the two longer sides carry one tick, the two shorter sides carry two ticks),
• it has two pairs of parallel sides,
• all its angles are equal (90° each),
• it has two lines of symmetry,
• its diagonals bisect it, but not at a right angle.

Notice the similarity: both are quadrilaterals whose angles are all 90°. The difference is that a square's four sides are all equal, while a rectangle only needs its opposite sides equal.

Visual



Squares Rectangles

Square: a right-angle mark at each corner, a single tick on each of the four sides. Rectangle: a right-angle mark at each corner, a single tick on each of the two longer sides and a double tick on each of the two shorter sides.

Now the parallelogram and the trapezium, in fig-03. The arrows on the sides mark which sides are parallel: one arrow means one pair of parallel sides, two arrows mark the other pair.

The parallelogram:
• its two pairs of opposite sides are equal and parallel,
• opposite angles are equal,
• it has no line of symmetry,
• its diagonals divide it into equal triangles.

The trapezium:
• only a pair of sides are parallel,
• opposite angles are not equal,
• it has no line of symmetry,
• its diagonal divides it into two unequal triangles.

Here is the key difference: a parallelogram has two pairs of parallel sides, while a trapezium has only one pair.

Visual



Parallelogram Trapezium

Parallelogram: a single arrow on each of the two horizontal sides, a double arrow on each of the two slanted sides. Trapezium: a single arrow on each of the two parallel sides, and one diagonal drawn in.

Finally, a polygon is a closed shape with three or more straight sides. Each polygon has a name that comes from its number of sides, and every polygon with n sides can be divided into n − 2 triangles by drawing diagonals from one vertex. Here is the list to learn:

Name of polygon	Number of sides	Number of triangles (n − 2)
Triangle	3	1
Quadrilateral	4	2
Pentagon	5	3
Hexagon	6	4
Heptagon	7	5
Octagon	8	6
Nonagon	9	7
Decagon	10	8
Hendecagon (undecagon)	11	9
Dodecagon	12	10

The rule n − 2 works for every polygon. A hexagon has 6 sides, so it splits into 6 − 2 = 4 triangles; a dodecagon has 12 sides, so it splits into 12 − 2 = 10 triangles.

Practice

1. How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices?
2. Name the triangle in which all three sides are equal, and state the size of each of its angles.
3. A shape has two pairs of parallel sides, all its angles are 90° each, but only its opposite sides are equal. Is it a square or a rectangle? Explain.
4. State one difference between a parallelogram and a trapezium.
5. A nonagon has how many sides, and into how many triangles can it be divided?

Section 2: Parts of a Circle

A circle is a special kind of plane shape, and to talk about it clearly we give names to its different parts. Look at the two labelled circles in fig-04 as you read.

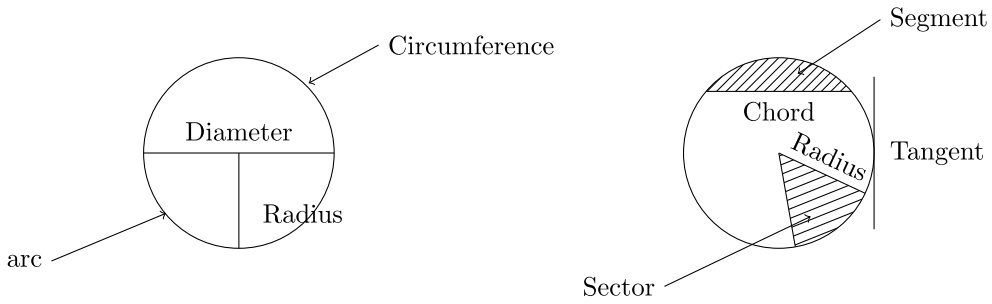
The **circumference** is the curved boundary of the circle — the round line that goes all the way around it. Any curved part of that circumference is called an **arc**.

Inside the circle, draw a straight line from the very centre of the circle to the circumference. That line is the **radius** (more than one are called radii). Every radius of the same circle is the same length.

Now draw a straight line joining any two points on the circumference. That line is called a **chord**. A chord cuts across the inside of the circle. When a chord passes through the centre of the circle, it is given a special name — the **diameter**. So the diameter is just a chord that goes through the centre, and it is the longest chord of the circle. Notice that a diameter is two radii put end to end.

Two more parts are shown in the second circle. The region between a chord and the arc above it is called a **segment** (in fig-04 it is the shaded part above the chord). The region between two radii and the arc between them — a slice of the circle, like a piece of cake — is called a **sector**. Finally, a **tangent** is a straight line that touches the circle at exactly one point and does not cut across it.

Visual



Two labelled circles. The first has the diameter drawn across it and a radius from its centre to the circumference, with the arc and the circumference labelled. The second has a chord near the top with the segment above it shaded, a shaded sector below with its right-hand bounding line labelled Radius, and a tangent touching it on the right.

Practice

1. A straight line is drawn from the centre of a circle to a point on the circumference. What is this line called?
2. Which part of a circle is a straight line joining two points on the circumference that passes through the centre?
3. What do we call a curved part of the circumference of a circle?
4. A straight line touches a circle at exactly one point without crossing it. What is it called?
5. A diameter of a circle is 8 cm long. How long is the radius of that circle?

Section 3: The Perimeter of Plane Shapes

The perimeter of a shape is the distance all the way round it. To find it, you add up the lengths of all the sides. Because you already know angles, you can recognise each shape's special feature that makes finding its perimeter quicker: a square has four equal sides, a rectangle has two equal lengths and two equal breadths, and so on.

Square. A square has four equal sides. If one side is L, the perimeter is the sum of all four:

$$L + L + L + L = 4L$$

So the perimeter of a square is $4 \times$ (length of one side).

Example 1. Calculate the perimeter of a square of side 15cm.

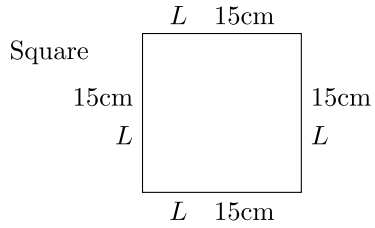
fig-05 shows a square with all four sides 15cm.

Since the four sides are equal,

$$\text{Perimeter} = L + L + L + L = 4L$$

\[\phantom{\text{Perimeter}} = 15 + 15 + 15 + 15 = 4(15)\]
\[\phantom{\text{Perimeter}} = 60\mathrm{cm}\]
So the square's perimeter is 60cm.

Visual



A square, named *Square* at its left, with all four sides marked twice — L and 15cm.

Rectangle. A rectangle has two equal lengths L and two equal breadths b. Adding all four sides:

$$L + b + L + b = 2L + 2b = 2(L + b)$$

So the perimeter of a rectangle is twice the sum of its length and breadth.

Example 2. Find the perimeter of a rectangle whose length is 10cm and breadth 7cm.

fig-06 shows a rectangle with length $L = 10\text{cm}$ and breadth $b = 7\text{cm}$.

Adding the four sides:

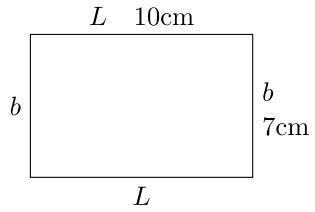
$$L + b + L + b = 10 + 7 + 10 + 7$$

Rearranging,

$$= 10 + 10 + 7 + 7 = 2(10) + 2(7)$$
$$= 2(10 + 7) = 2(17)$$
$$= 34\text{cm}$$

The rectangle's perimeter is 34cm.

Visual



A rectangle with both long sides marked L, the top one also 10cm, and both short sides marked b, the right-hand one also 7cm.

Example 3. If the perimeter of a rectangle is 60cm and its length is 17cm, calculate its breadth.

fig-07 shows a rectangle with length $L = 17\text{cm}$ and unknown breadth b.

Using $\text{Perimeter} = L + b + L + b$,

$$60 = 17 + b + 17 + b$$
$$60 = 17 + 17 + b + b = 2(17) + 2b$$
$$60 = 34 + 2b$$

Collect the like terms:

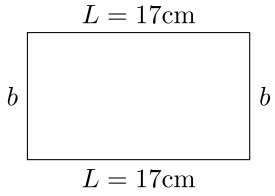
$$60 - 34 = 2b$$
$$2b = 26$$

Divide both sides by 2:

$$\frac{2b}{2} = \frac{26}{2}$$
$$b = 13\text{cm}$$

The breadth of the rectangle is 13cm.

Visual



A rectangle with both long sides marked $L = 17$cm and both short sides marked b.

Trapezium. A trapezium has four sides of no particular equal lengths, so its perimeter is simply the sum of all four sides.

Parallelogram. A parallelogram has two pairs of equal sides: two equal sides a and two equal sides b. Its perimeter is therefore

$$a + a + b + b = 2a + 2b = 2(a + b)$$

Example 4. Find the perimeter of the trapezium below.

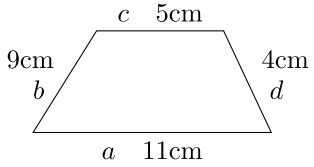
fig-08 shows a trapezium with sides $a = 11$cm, $b = 9$cm, $c = 5$cm and $d = 4$cm.

Adding the four sides:

$$\begin{aligned}\text{Perimeter} &= a + b + c + d \\ &= 11 + 9 + 5 + 4 \\ &= 29\text{cm}\end{aligned}$$

The trapezium's perimeter is 29cm.

Visual



A trapezium with each side lettered and measured: the top c, 5cm; the left b, 9cm; the right d, 4cm; the base a, 11cm.

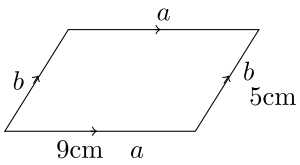
Example 5. Find the perimeter of the parallelogram below.

fig-09 shows a parallelogram with sides $a = 9$cm and $b = 5$cm, each side occurring twice.

$$\begin{aligned}\text{Perimeter} &= a + a + b + b = 2a + 2b = 2(a + b) \\ &= 2(9 + 5) = 2(14) \\ &= 28\text{cm}\end{aligned}$$

The parallelogram's perimeter is 28cm.

Visual



A parallelogram with a single arrow on each of its four sides: the two horizontal ones both lettered a, the lower also measured 9cm, and the two slanted ones both lettered b, the right-hand one also measured 5cm.

Regular polygon. A regular polygon has all its sides equal. A regular hexagon has 6 sides, a regular pentagon has 5 sides. So the perimeter of a regular polygon is

$$\text{(number of sides)} \times \text{(length of one side)}.$$

Example 6. Find the perimeter of a regular hexagon with sides 5.2cm.

fig-10 shows a regular hexagon with all six sides 5.2cm.

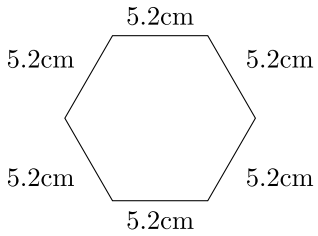
A hexagon has 6 sides, all equal:

$$\begin{aligned}\text{Perimeter} &= 6 \times 5.2\text{cm} \\ &= 31.2\text{cm}\end{aligned}$$

The hexagon's perimeter is 31.2cm.

Visual

Regular hexagon



A regular hexagon, named above the drawing, with each of its six sides marked 5.2cm.

Circle. The perimeter of a circle has its own name: the circumference. If the circle has radius r and diameter $d = 2r$, then

$$\text{Circumference} = 2\pi r \quad \text{or} \quad \pi d.$$

You can use either formula; choose the one that fits what you know.

Example 7. Calculate the perimeter of a circle of radius 7cm. Use $\pi = \frac{22}{7}$.

fig-11 shows a circle with radius 7cm.

Using $\text{Perimeter} = 2\pi r$:

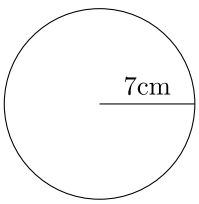
$$\begin{aligned}&= 2 \times \tfrac{22}{7} \times 7 \\ &= 2 \times 22 \\ &= 44\text{cm}\end{aligned}$$

We can check with πd. Since $r = 7$cm, the diameter $d = 2 \times 7 = 14$cm, and

$$\tfrac{22}{7} \times 14 = 22 \times 2 = 44\text{cm}.$$

The circumference is 44cm.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked 7cm.

Example 8. The circumference of a circle is 440cm. Find its radius. Use $\pi = \frac{22}{7}$.

fig-13 shows a circle with unknown radius r.

Using $\text{Circumference} = 2\pi r$:

$$440 = 2 \times \tfrac{22}{7} \times r$$
$$440 = \tfrac{44r}{7}$$

Cross multiply:

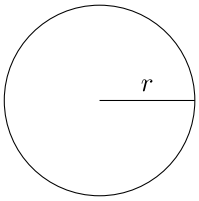
$$440 \times 7 = 44r$$
$$3080 = 44r$$

Divide both sides by 44:

$$\tfrac{44r}{44} = \tfrac{3080}{44}$$
$$r = 70\text{cm}$$

The radius of the circle is 70cm.

Visual



A circle with a radius drawn from the centre to the right of the circumference, marked r.

Example 9. A rope is wound 50 times round a cylinder of radius 25cm. How long is the rope? Use $\pi = 3.14$.

fig-12 shows the rope wound round the cylinder. One complete turn of the rope is the circumference of the cylinder, so the length of rope in one turn is the circumference.

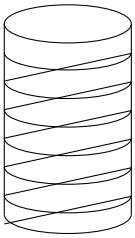
$$\begin{aligned}\text{Circumference} &= 2\pi r = 2 \times 3.14 \times 25 \\ &= 157\text{cm}\end{aligned}$$

The rope goes round 50 times, so the total length of rope is

$$157\text{cm} \times 50 = 7850\text{cm}.$$

The rope is 7850cm long.

Visual



An upright cylinder with a rope wound round it, drawn as a series of turns spiralling down the body from the top face to the base.

Combined shapes. Some shapes are made of straight sides and a curved arc. For these, find the length of the curved part and add it to the straight sides. A quarter circle is one quarter of a full circle, so its curved length is

$$\tfrac{1}{4} \times 2\pi r.$$

Example 10. Calculate the perimeter of the shape below.

fig-14 shows a quarter circle: an upright side and a base, each 4.9cm, joined by an arc.

First find the length of the curved part. The arc is a quarter of a circle of radius 4.9cm:

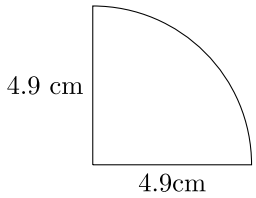
$$\begin{aligned}\text{curved length} &= \tfrac{1}{4} \times 2\pi r = \tfrac{1}{4} \times 2 \times \tfrac{22}{7} \times 4.9 \\ &= \tfrac{1}{4} \times \tfrac{44}{7} \times 4.9 = \tfrac{11}{7} \times \tfrac{49}{10} \\ &= \tfrac{11 \times 7}{10} = \tfrac{77}{10} = 7.7\text{cm}\end{aligned}$$

Now add the two straight sides:

$$\text{Perimeter} = 4.9 + 4.9 + 7.7 = 17.5\text{cm}$$

The perimeter of the shape is 17.5cm.

Visual



A quarter circle: an upright side and a base meeting at a right angle, each marked 4.9cm, with an arc closing them from the top of the upright to the end of the base.

Practice

1. Find the perimeter of a square of sides 8cm.
2. Find the perimeter of a rectangle of length 7cm and breadth 4cm.
3. What is the perimeter of a regular pentagon with sides 5cm?
4. Find the perimeter of a parallelogram whose base is 6cm and sides 5cm.
5. The circumference of a circle is 22cm. Find the radius of the circle. Use $\pi = \frac{22}{7}$.
6. A rectangle is such that its length is twice its breadth. If the breadth is d, find the total length of the sides of the rectangle.
7. Find the perimeter of a regular hexagon with sides 5cm.
8. Calculate the perimeter of the figure below: a base of 9cm, a right side of 5cm, and a top made of two 1cm straight pieces joined by a semicircular arc. Use $\pi = \frac{22}{7}$.

Check your work

Attempt every question before you read this page.

Types of Plane Shapes and Their Properties

1. 6

   An octagon has 8 sides, so n = 8. Using n − 2 gives 8 − 2 = 6 triangles.
2. Equilateral triangle; each angle is 60°.
3. It is a rectangle, because a square has all four sides equal, while a rectangle only has its opposite sides equal.
4. A parallelogram has two pairs of parallel sides, while a trapezium has only one pair of parallel sides.
5. 9 sides; 7 triangles (9 − 2 = 7).

Parts of a Circle

1. Radius.
2. Diameter.
3. An arc.
4. A tangent.
5. 4 cm.

   The diameter is a chord through the centre and is two radii put end to end. So the radius is half the diameter: 8 cm ÷ 2 = 4 cm.

The Perimeter of Plane Shapes

1. 32cm

   Perimeter of a square $= 4L = 4 \times 8\text{cm} = 32\text{cm}$.
2. 22cm

   $\text{Perimeter} = 2(L + b) = 2(7 + 4) = 2(11) = 22\text{cm}$.

3. 25cm

   A regular pentagon has 5 equal sides, so Perimeter $= 5 \times 5\text{cm} = 25\text{cm}$.
4. 22cm

   The parallelogram has two sides of 6cm and two of 5cm: Perimeter $= 2(6+5) = 2(11) = 22\text{cm}$.
5. 3.5cm

   $22 = 2 \times \frac{22}{7} \times r = \frac{44r}{7}$. Cross multiplying, $22 \times 7 = 44r$, so $154 = 44r$ and $r = \frac{154}{44} = 3.5\text{cm}$.
6. $6d$

   Length $= 2d$ and breadth $= d$, each occurring twice. So the total length of the sides is $2(2d) + 2(d) = 4d + 2d = 6d$.
7. 30cm

   A regular hexagon has 6 equal sides, so Perimeter $= 6 \times 5\text{cm} = 30\text{cm}$.
8. 27cm

   The semicircular arc has diameter 7cm, so radius 3.5cm, and its length is $\pi r = \frac{22}{7} \times 3.5 = 11\text{cm}$.

   Now add the straight sides: base 9cm, right side 5cm, and the two 1cm top pieces:

   $$9 + 5 + 1 + 1 = 16\text{cm}$$

   Total perimeter $= 16 + 11 = 27\text{cm}$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-8a7a98fa	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____